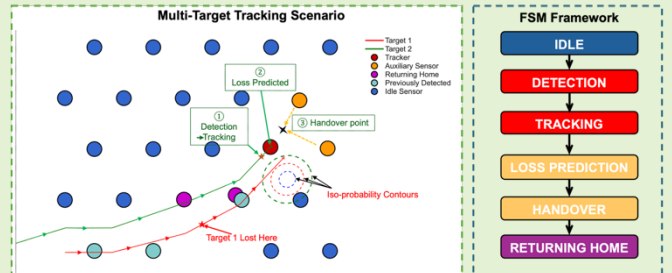


Reconfigurable Wireless Sensor Network Coordination for Simultaneous Multi-Target Tracking

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Abstract— This paper presents a distributed coordination framework for simultaneous multi-target tracking using a wireless sensor network based on discrete-event-system principles. The proposed framework employs a finite-state-machine architecture, where autonomous mobile sensors sequentially process detection and tracking events. Unlike passive tracking approaches that rely on single-sensor monitoring, our proposed strategy implements predictive handover through Extended Kalman Filter based uncertainty propagation, enabling sensors to anticipate target loss events and proactively coordinate auxiliary sensors to reposition along predicted trajectories using navigation guidance. A bidding-based allocation mechanism coordinates sensor assignments by evaluating four competing objectives: network preservation, spatial proximity to handover points, temporal mission feasibility, and estimation uncertainty. The key novelty lies in integrating predictive loss detection with distributed competitive bidding for multi-target scenarios within a reconfigurable sensor network. This approach eliminates a need for centralized control, while maintaining network topology. Extensive simulation results demonstrate tangible improvement in tracking coverage and sensor engagement compared to reactive tracking baselines, across scenarios with varying target speeds and trajectories.

Index Terms— Multi-target tracking, wireless sensor networks, discrete-event-system control, navigation guidance, predictive tracking, distributed coordination



I. INTRODUCTION

AUTONOMOUS simultaneous tracking of multiple mobile targets represents a fundamental challenge in distributed robotic systems with applications spanning surveillance [1], search and rescue [2–4], environmental monitoring [5], etc. Dynamic wireless sensor networks (WSNs) offer a scalable solution for effective area coverage, where spatially distributed sensors can cooperate to detect and track targets moving through the monitored region [6,7]. However, maintaining continuous target observation in such environments remains problematic when individual sensors possess limited detection ranges and mobility constraints that restrict their operational displacement from designated home positions [8].

Traditional WSN tracking approaches suggest passive handover strategies where sensors track targets individually until detection is lost and, then, rely on opportunistic encounters with neighboring sensors to reacquire the target [9], [10]. This reactive paradigm results in frequent target loss events, particularly when target velocities exceed sensor

mobility capabilities or when targets traverse sparsely covered regions [11]. A passive approach may fail to leverage the distributed sensing and communication capabilities inherent in networked systems, treating the WSN as a collection of independent agents rather than a coordinated system [12]. Furthermore, such methods may fail to predict when current trackers will lose target observation, preventing proactive positioning of additional sensors to ensure continuous tracking [13], [14].

This paper presents a novel distributed coordination framework that implements predictive active handover, where sensors use Extended Kalman Filter (EKF) based uncertainty propagation to anticipate tracker loss events and proactively position auxiliary sensors along predicted target trajectories using guidance-theory principles [15], [16]. The proposed discrete-event-system (DES) architecture coordinates autonomous sensors through finite-state machines (FSMs), enabling dynamic role transitions between detecting and tracking based on target behavior and network conditions.

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The main contributions of this work are: (1) a predictive handover mechanism using EKF convergence analysis to trigger loss prediction and auxiliary sensor deployment based on guidance theory, (2) a multi-objective bidding mechanism balancing network preservation, spatial proximity, temporal feasibility, and estimation uncertainty for distributed auxiliary sensor allocation, (3) a conflict resolution strategy ensuring discrete event compliance when multiple targets simultaneously require resources, and (4) lightweight EKF-based iso-probability contour generation enabling real-time uncertainty representation for future search operations.

The remainder of this paper is organized as follows. Section II reviews related work in multi-target tracking. Section III formulates the problem. Section IV presents the proposed DES architecture. Section V includes simulation results demonstrating tracking performance. Section VI concludes with discussion of key findings and future research directions.

II. RELATED WORK

A. Predictive Target Tracking

Multi-target tracking in distributed sensor networks has evolved from reactive detection-based approaches to predictive frameworks that anticipate tracking failures before they occur. Active tracking algorithms in [17] include provable approximation bounds for handling sensor failures and occlusions, demonstrating that proactive failure detection outperforms reactive recovery. Distributed tracking architectures were extended to autonomous vehicle fleets in [18] and heterogeneous mobile sensing networks in [19], where normalized unused sensing capacity metrics enable automatic load balancing to prevent coverage gaps.

Directional-sensor network deployment strategies [20] and dynamic deployment approaches for expanding search areas [21] enable adaptive coverage optimization. Hybrid strategies combining static and mobile sensors [22] with spatiotemporal optimization [23] improve target detection through coordinated sensing and adaptive deployment.

The challenge of maintaining continuous observation with limited sensing ranges has also motivated research in collision-free tracking with localization uncertainty [24] and interactive Kalman filtering for multi-vehicle coordination [25]. Distributed state estimation frameworks address tracking loss prediction through modified unscented Kalman filters (KFs) in fading channel environments [26] and simultaneous sensor position refinement [27]. Consensus-based mixture KFs [28] enable tracking of maneuvering targets by predicting conditional dynamics changes, while multi-target information consensus algorithms [29] specifically address tracking loss in non-overlapping sensor fields of view.

Recent advances in vision-based reactive planning [30]

incorporate predictive horizons for occlusion handling, and heterogeneous limited-range sensor frameworks [31] use cooperative coverage prediction to prevent tracking gaps. However, these approaches primarily focus on state estimation quality rather than explicitly predicting when and where current trackers will lose targets, creating opportunities for proactive sensor repositioning based on loss prediction rather than reactive reacquisition.

B. Distributed Coordination and Task Allocation

Task allocation in multi-robot systems has transitioned from centralized optimization to distributed mechanisms enabling scalable coordination without global knowledge. Group-based distributed auction algorithms [32] address time-window constraints and conflict resolution through competitive bidding, while factor graph formulations with max-sum algorithms [33] enable fully decentralized task allocation. Resource-oriented auction mechanisms [34] consider energy constraints and limited communication ranges, demonstrating that distributed bidding outperforms centralized allocation in communication-constrained environments. Active-vision-based multi-sensor surveillance systems [35] demonstrate effective coordination for multi-target monitoring in dynamic environments.

Network topology preservation has also received attention through information space feedback control [36] and multi-robot coordination with task precedence relationships [37]. Conflict resolution, when multiple tasks compete for limited resources, has been addressed through metaheuristic optimization [38] and hierarchical FSMs [39]. Connectivity recovery methods [40] use FSM-based distributed coordination to identify critical nodes and coordinate network self-healing, while consensus-based bundle algorithms [41] enable task allocation in teams with severely limited communication.

Reconfigurable coordination frameworks [42] provide virtual coordinator architectures for discrete-event control systems, enabling dynamic role transitions without centralized supervisors. However, current task-allocation methods typically optimize a single objective or use fixed-priority schemes, lacking multi-objective frameworks that simultaneously balance network topology preservation, spatial mission costs, temporal feasibility, and information quality which are all critical for maintaining persistent tracking coverage while minimizing network disruption.

C. Target Pursuit

Guidance theory has been successfully adapted to robotic systems for target interception and pursuit-evasion scenarios. Navigation-guidance-based approaches [43] and augmented proportional navigation techniques [44] have demonstrated effective robotic pursuit of moving objects. Building on these foundations, proportional navigation guidance (PNG) has

been adapted for robotic target pursuit with final velocity control [45], enabling smooth rendezvous with moving targets.

Optimal guidance strategies for UAV swarms [46] integrate pure pursuit laws with formation maintenance, while bio-inspired cooperative hunting behaviors [47] demonstrate mixed guidance approaches using Markov decision processes for multi-agent coordination. Pursuit-evasion frameworks have been developed for multi-robot systems with limited sensing [48], incorporating evolutionary robotics approaches for decentralized control. Three-dimensional guidance laws [49] extend proportional navigation to rotary UAVs with velocity feedback under stochastic conditions. Asymptotically-optimal visibility-based pursuit-evasion [49] uses hierarchical graph structures for multi-robot coordination, while fair multi-target tracking frameworks [50] balance target assignment fairness with tracking performance.

DES formulations have been applied to incremental scheduling [51] and reconfigurable coordination [42] in multi-robot systems, providing formal frameworks for event-driven control. However, past guidance-based approaches for robotics focus primarily on direct pursuit or rendezvous problems rather than proactive handover coordination to prevent tracking loss in sensor networks. The integration of guidance theory with distributed sensor coordination for predictive handover - where sensors position themselves at predicted handover points before current trackers lose targets - remains largely unexplored in WSN multi-target tracking applications.

III. PROBLEM FORMULATION

This section formulates the multi-target tracking problem for reconfigurable wireless sensor networks. We consider a system of mobile sensors with limited detection ranges tracking multiple targets with unknown trajectories. The objective is to maximize target tracking coverage while minimizing sensor displacement from designated home positions.

A. System Model and Assumptions

Let us consider a wireless sensor network (WSN) comprising N mobile sensors deployed in an optimal (herein, hexagonal) grid configuration with inter-node spacing d . Although the system may operate in a three-dimensional environment, for example, utilizing a team of UAVs, the third dimension is omitted herein. Namely, only the planar placement of sensors is investigated.

All parameters (positions, distances, ranges) are expressed in arbitrary spatial units, with temporal quantities in corresponding time unit. The notation $|\cdot|$ denotes the Euclidean norm ($L2$ distance) in two-dimensional space.

Each sensor, $i = 1 \dots N$, has a designated home position

$\mathbf{h}_i \in \mathbb{R}^2$ and current position $\mathbf{s}_i(t) \in \mathbb{R}^2$, where positions are two-dimensional vectors representing planar coordinates. Sensors possess identical capabilities: detection radius, r_d , maximum velocity, v_s , and communication range, r_c .

To preserve network topology, each sensor is constrained to operate within a maximum distance, d_{max} , from its home position. Multiple targets, $i = 1 \dots K$, enter the monitored region with unknown entry times and locations. Each target follows a piecewise linear trajectory defined by waypoints $W_k = \{\omega_1 \omega_2 \dots \omega_m\}$, moving with constant velocity $\mathbf{v}_k$ between consecutive waypoints. The state of target k at time t is represented as:

$$\mathbf{x}_k(t) = [\mathbf{p}_k(t) \mathbf{v}_k(t)]^T \quad (1)$$

where $\mathbf{p}_k(t) = [x_k \ y_k]^T$ denotes the target position and $\mathbf{v}_k(t) = [v_k^x \ v_k^y]^T$ denotes the velocity vector.

Sensor i detects target k when $|\mathbf{s}_i(t) - \mathbf{p}_k(t)| \leq r_d$. Upon detection, the sensor obtains noisy position measurements:

$$\mathbf{z}_{ik}(t) = \mathbf{p}_k(t) + \mathbf{n}(t) \quad (2)$$

where $\mathbf{n}(t) \sim \mathcal{N}(\mathbf{0} \ \mathbf{R})$ represents Gaussian measurement noise with covariance $\mathbf{R}$.

B. Problem Statement

Given N sensors with home positions $\{\mathbf{h}_i\}_{i=1}^N$ and K targets with trajectories $\{\mathbf{x}_k(t)\}_{k=1}^K$, the problem at hand is to maximize target tracking coverage while minimizing total sensor displacement from home positions, expressed as:

$$\text{minimize } J = \sum_{i=1}^N \|\mathbf{s}_i(t) - \mathbf{h}_i\|^2 \quad (3)$$

subject to the following constraints:

- (1) **Target Coverage:** The system aims to maximize the time each target is detected by at least one sensor; specifically, maximizing the proportion of time where $|\mathbf{s}_i(t) - \mathbf{p}_k(t)| \leq r_d$ for at least one sensor while the target remains within the monitored region. Continuous coverage cannot be guaranteed when targets move faster than sensors ($|\mathbf{v}_k| > v_s$) or when targets approach region boundaries where sensor displacement constraints prevent adequate coverage.
- (2) **Displacement Constraint:** Each sensor must respect the maximum displacement constraint: $|\mathbf{s}_i(t) - \mathbf{h}_i| \leq d_{max}$, which preserves network topology and limits energy consumption.
- (3) **Measurement Noise:** Tracking decisions must account for noisy position measurements $\mathbf{z}_{ik}(t)$ to determine target velocity $\mathbf{v}_k$ and predict future positions.

C. Sub-Problems

To address the overall problem stated above, four critical

sub-problems must be solved: tracking loss prediction, proactive positioning, distributed sensor allocation, and conflict resolution. They are detailed below:

- (1) **Tracking Loss Prediction:** Determine when current tracking sensors will lose target observation, given limited detection range, r_d , and target mobility. Specifically, for each tracking sensor and target, the predicted loss time τ must be determined such that:

$$|\mathbf{s}_i(t + \tau) - \mathbf{p}_k(t + \tau)| > r_d \quad (4)$$

This computation must account for: (i) target velocity uncertainty arising from noisy measurements, (ii) sensor pursuit capability constraints, when $|\mathbf{v}_k| > v_s$, and (iii) state-estimation convergence requirements for reliable prediction.

- (2) **Proactive Positioning:** Determine the repositioning locations for auxiliary sensors before tracking loss occurs. Specifically, predict handover points, $\mathbf{p}_{\text{handover}}$, where auxiliary sensors can acquire targets before current trackers lose detection. The calculations must account for sensor travel time and communication delays to ensure auxiliary sensors arrive with sufficient temporal margin.
- (3) **Distributed Sensor Allocation:** Determine which sensors should be allocated to handover tasks, while balancing competing objectives: (i) network topology preservation, by minimizing $|\mathbf{s}_i - \mathbf{h}_i|$ to maintain the designed WSN configuration, (ii) spatial efficiency, by minimizing distance to handover points, (iii) temporal feasibility, by ensuring arrival before predicted target loss, and (iv) information quality, by selecting sensors with low tracking uncertainty. The allocation must be determined in a distributed manner without centralized control, with each sensor evaluating its own suitability based on local information.
- (4) **Conflict Resolution:** Resolve resource allocation when multiple targets require auxiliary sensor assistance while prior handover requests are still being processed. Specifically, determine: (i) when handover calls from different targets occur within overlapping time windows, (ii) how to allocate non-overlapping auxiliary sensor teams to competing targets, and (iii) how to ensure no target is disproportionately disadvantaged in resource allocation.

IV. PROPOSED APPROACH

A. System Architecture and Configuration

As noted above, for simplicity, it is assumed herein that the optimal WSN configuration is a $n_{\text{grid}} \times n_{\text{grid}}$ hexagonal grid with $N = n_{\text{grid}}^2$ sensors and inter-node spacing d . While our solution framework allows for any optimal network configuration, we select the hexagonal topology as it provides

optimal two-dimensional coverage without directional bias, minimizing the number of sensors required for complete area monitoring [52], [53], [54]. The staggered hexagonal geometry simplifies distributed coordination algorithms and enables predictable communication patterns.

As shown in Fig. 1, the displacement constraint is set to $d_{\text{max}} = d$, ensuring sensors remain within communication range, r_c , of their neighbors, while providing sufficient mobility for target interception. With communication range $r_c > d$, sensors at home positions maintain connectivity with their neighbors. This connectivity is preserved during repositioning as the constraint $d_{\text{max}} = d$ prevents sensors from moving beyond communication range of the network. Grid parameters balance sensor utilization: larger spacing d reduces redundancy but risks coverage gaps, while smaller spacing increases robustness at the cost of sensor density.

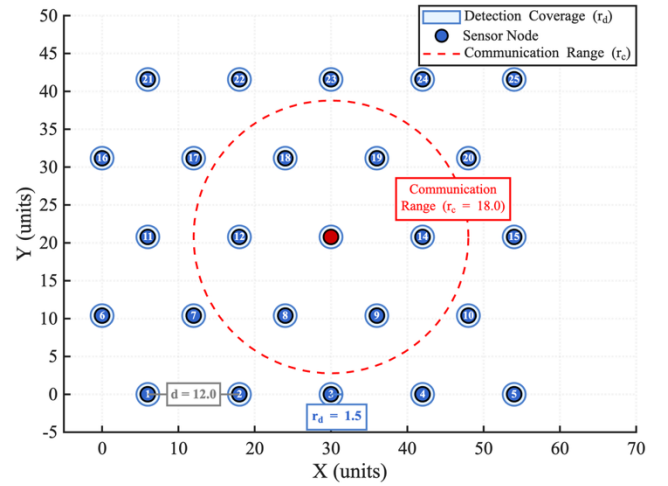


Fig. 1 Hexagonal-grid WSN deployment

B. DES Architecture and Coordination Framework

The proposed framework addresses the four sub-problems identified in Section III.C through a discrete-event-system (DES) architecture. Specifically:

- *Sub-problem C.1* (Tracking Loss Prediction) is solved through EKF-based uncertainty propagation (Sections IV.C-D)
- *Sub-problem C.2* (Proactive Positioning) is solved through predictive handover point computation and adaptive guidance (Sections IV.D, IV.H)
- *Sub-problem C.3* (Distributed Sensor Allocation) is solved through multi-objective competitive bidding (Section IV.E)
- *Sub-problem C.4* (Conflict Resolution) is solved through min-max optimization for simultaneous requests (Section IV.F)

Supporting mechanisms include information sharing between neighboring sensors (Section IV.G), dual-tracker configuration for robustness (Section IV.I), and EKF-based search contour generation for post-loss reacquisition (Section

IV.J).

The framework operates as a DES where state transitions occur in response to well-defined events rather than continuous time updates. This architecture enables efficient coordination by processing discrete tracking events (target detections, predicted losses, handover requests), while minimizing computational overhead during periods of tracking.

Each sensor operates according to a finite-state machine (Fig. 2) with six states $S = \{IDLE, DETECTING, TRACKING, REPOSITIONING, RETURNING_HOME\}$. State transitions are event-driven:

- $IDLE \rightarrow DETECTING$ occurs upon target detection within range
- $DETECTING \rightarrow TRACKING$ occurs when fewer than two sensors are currently tracking the target (dual-tracker configuration has available slots)
- $IDLE \rightarrow REPOSITIONING$ occurs when an idle sensor wins bidding for a handover call
- $REPOSITIONING \rightarrow TRACKING$ occurs upon reaching the predicted handover point and successfully acquiring the target
- $TRACKING \rightarrow RETURNING_HOME$ occurs when the sensor loses target detection
- $REPOSITIONING \rightarrow RETURNING_HOME$ occurs when the sensor cannot acquire the target at the handover point
- $Any\ state \rightarrow RETURNING_HOME$ occurs when the sensor exceeds the displacement threshold d_{max} without active mission.

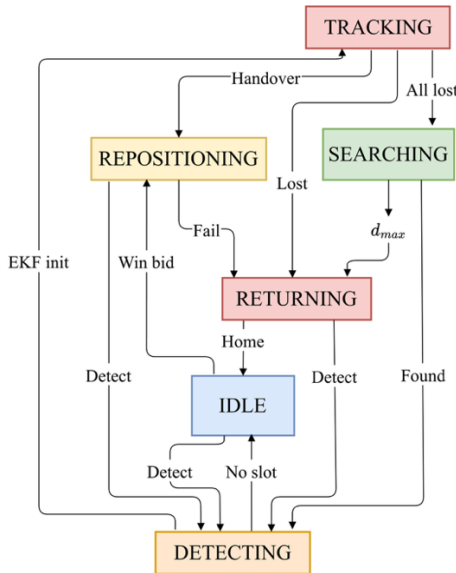


Fig. 2 Proposed sensor finite-state-machine architecture.

The proposed architecture integrates three coordinating modules via message passing (Fig. 3): the Tracking

Coordinator manages tracker assignments and immediate handover requests, the Handover Coordinator processes predictive handover calls through distributed bidding and conflict resolution, and the Recovery Protocol manages return-to-home operations and generates iso-probability contours for future search operations. This separation enables parallel processing of independent tracking tasks while ensuring coordinated resolution of resource conflicts.

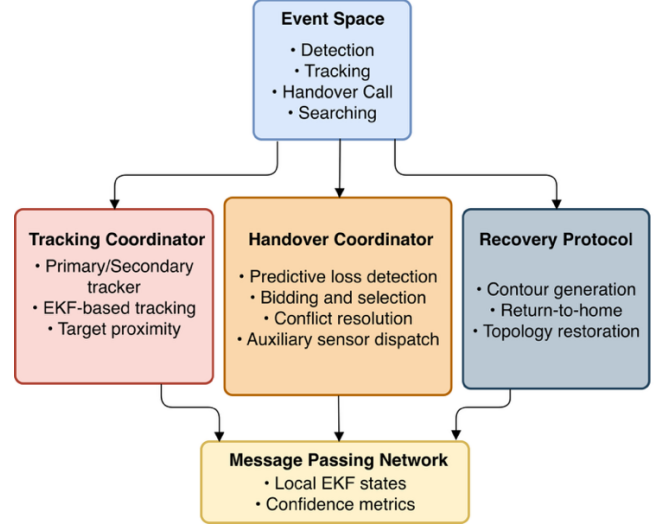


Fig. 3 DES architecture.

Importantly, decision-making is distributed: each sensor executes local decisions based on its finite state machine (Fig. 2) using information from coordinating modules. The system employs a hybrid communication model where global broadcast enables all sensors to participate in competitive bidding for handover coordination, while local peer-to-peer communication within range r_c handles routine tracking information exchange (detailed in Section IV.G).

C. EKF-Based Target State Estimation

To solve Sub-problem C.1 (Tracking Loss Prediction), each tracking sensor maintains an EKF estimating target state $\hat{\mathbf{x}}_k(t)$ with covariance $\mathbf{P}_k(t)$. The discrete-time state transition model with time step Δt is:

$$\mathbf{x}_k(t + \Delta t) = \mathbf{F}\mathbf{x}_k(t) + \mathbf{G}\mathbf{w}(t) \quad (5)$$

where $\mathbf{F}$ is the state transition matrix, $\mathbf{G}$ is the process noise gain, and $\mathbf{w}(t) \sim \mathcal{N}(0, \mathbf{I})$ represents unit process noise. The matrices are defined as:

$$\mathbf{F} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \Delta t^2/2 & 0 \\ 0 & \Delta t^2/2 \\ \Delta t & 0 \\ 0 & \Delta t \end{bmatrix} \quad (6)$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{Q} = \sigma_w^2 \mathbf{I}_2$$

where σ_w^2 represents process noise covariance. The EKF

prediction step propagates state and covariance:

$$\hat{\mathbf{x}}_k^- = \mathbf{F}\hat{\mathbf{x}}_{k-1} \quad \mathbf{P}_k^- = \mathbf{F}\mathbf{P}_{k-1}\mathbf{F}^T + \mathbf{G}\mathbf{Q}\mathbf{G}^T \quad (7)$$

Upon receiving measurement $\mathbf{z}_{ik}$, the update step determines the Kalman gain $\mathbf{K}$ and updates:

$$\mathbf{K} = \mathbf{P}_k^- \mathbf{H}^T (\mathbf{H}\mathbf{P}_k^- \mathbf{H}^T + \mathbf{R})^{-1} \quad (8)$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}(\mathbf{z}_{ik} - \mathbf{H}\hat{\mathbf{x}}_k^-) \quad \text{and} \quad \mathbf{P}_k = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}_k^-$$

Convergence is detected when velocity covariance components satisfy $\mathbf{P}_k(33) < \sigma_{v_{thresh}}^2$ and $\mathbf{P}_k(44) < \sigma_{v_{thresh}}^2$ for threshold $\sigma_{v_{thresh}}^2$ indicating sufficient velocity estimation quality for loss prediction.

D. Predictive Loss Detection Using EKF Convergence

Upon EKF convergence, tracking sensors address Sub-problem C.1 by predicting when target loss will occur. The sensor forward propagates the target state. Given tracker position $\mathbf{s}_i(t)$ and estimated target state $\hat{\mathbf{x}}_k(t) = [\hat{\mathbf{p}}_k(t) \ \hat{\mathbf{v}}_k(t)]^T$, the predicted distance at future time $t + \tau$ is:

$$d(\tau) = |\mathbf{s}_i(t) + \mathbf{v}_s \tau - (\hat{\mathbf{p}}_k(t) + \hat{\mathbf{v}}_k(t)\tau)| \quad (9)$$

for time offset τ .

When target velocity exceeds sensor capability $|\hat{\mathbf{v}}_k| > v_s$, loss is inevitable and τ_{loss} is determined by when the distance first exceeds r_d given sensor pursuit at maximum velocity toward the target, as illustrated in Fig. 4.

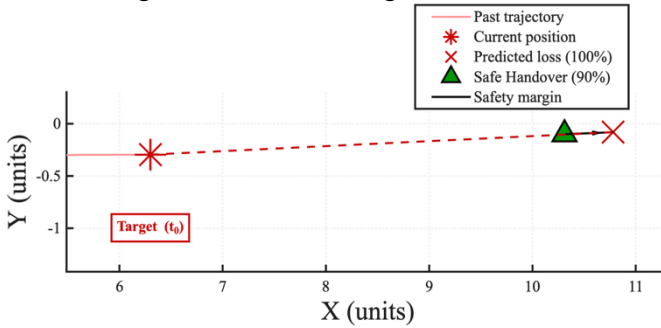


Fig. 4 Predictive loss detection and handover mechanism.

To ensure sensors arrive before tracking loss, a safety margin is applied: the handover point is set to $\mathbf{p}_{handover} = \hat{\mathbf{p}}_k(t) + \hat{\mathbf{v}}_k(t)\tau_{safe}$ where $\tau_{safe} < \tau_{loss}$ provides a temporal safety margin (greater values increase robustness but require earlier sensor commitment). Fig. 5 shows a detailed view of the safety margin application during the handover process.

E. Multi-Objective Bidding for Sensor Assignment

To solve Sub-problem C.3 (Distributed Sensor Allocation), we propose a competitive bidding mechanism where each available sensor evaluates its suitability for handover tasks. Upon receiving a handover call, each available Sensor i computes a bid reflecting four competing objectives. Let $\mathbf{s}_i$ denote current position, $\mathbf{h}_i$ home position, and $\mathbf{p}_{handover}$ the predicted handover point. The bid components are:

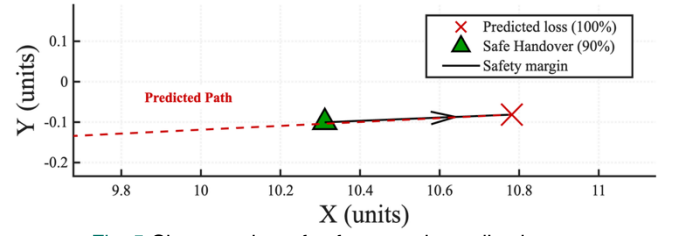


Fig. 5 Close up view of safety margin application.

(1) Network Topology Preservation (Objective i in Section III.C.3) quantifies displacement from home position:

$$\text{penalty}_{\text{network}}(i) = \frac{|\mathbf{s}_i - \mathbf{h}_i|}{D_{\max}} \quad (10)$$

where $D_{\max} = \sqrt{W^2 + H^2}$ is the network diagonal (maximum possible distance between any two points in the monitoring region of width W and height H).

(2) Spatial Efficiency (Objective ii in Section III.C.3) measures distance to handover point:

$$\text{penalty}_{\text{spatial}}(i) = \frac{|\mathbf{s}_i - \mathbf{p}_{handover}|}{D_{\max}} \quad (11)$$

(3) Temporal Feasibility (Objective iii in Section III.C.3) evaluates sensor arrival time relative to target arrival at handover point.

Let $t_{\text{sensor}} = |\mathbf{s}_i - \mathbf{p}_{handover}|/v_s$ be the sensor's time to reach the handover point, and

$$t_{\text{target}} = \frac{|\mathbf{p}_k(t) - \mathbf{p}_{handover}|}{|\mathbf{v}_k|} \quad (12)$$

be the target's time to reach the handover point. The temporal penalty is:

$$\text{penalty}_{\text{temporal}}(i) = \min \left(1, \max \left(0, \frac{t_{\text{sensor}} - t_{\text{target}}}{t_{\text{target}}} \right) \right) \quad (13)$$

This penalty is zero when the sensor arrives before the target (ensuring successful handover), but increases linearly with late arrival, and is capped at 1 to maintain comparable scales with other penalties.

(4) Information Quality (Objective iv in Section III.C.3) reflects the reliability of shared target information available to the repositioning sensor. Based on information age and spatial proximity to active trackers, each sensor computes a confidence metric $\alpha_i \in [0, 1]$ (see Section IV.G for details). The uncertainty penalty is:

$$\text{penalty}_{\text{uncertainty}}(i) = 1 - \alpha_i \quad (14)$$

where lower confidence in shared information increases the penalty, discouraging assignment when available information is stale or unreliable.

The total bid combines these components through a weighted summation:

$$\text{bid}(i) = w_n \cdot p_n(i) + w_s \cdot p_s(i) + w_t \cdot p_t(i) + w_u \cdot p_u(i) \quad (15)$$

where $p_n = \text{penalty}_{\text{network}}$, $p_s = \text{penalty}_{\text{spatial}}$, $p_t = \text{penalty}_{\text{temporal}}$, $p_u = \text{penalty}_{\text{uncertainty}}$, and $w_n w_s w_t w_u$ satisfy $\sum w = 1$. Sensors with lower bids are preferable.

As shown in Fig. 6, through consensus, the two sensors with lowest bids are selected, maintaining redundancy against prediction errors or sensor failures.

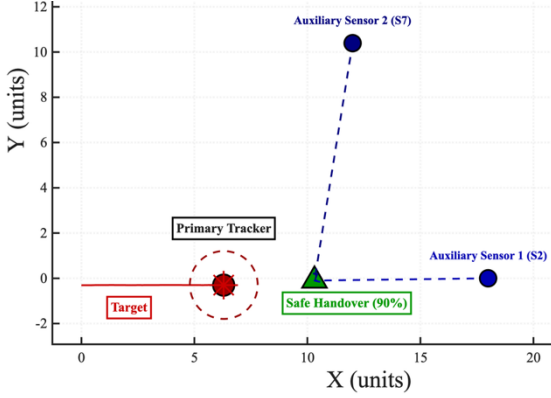


Fig. 6 Successful multi-objective bidding.

F. Conflict Resolution for Simultaneous Handover Requests

To solve Sub-problem C.4 (Conflict Resolution), a min-max optimization approach is proposed herein, when multiple targets simultaneously require auxiliary sensor assistance. Conflicts arise when handover calls from different tracking sensors occur within the same DES processing cycle. The system detects such conflicts when a second handover call is received while the first request is still being processed. Rather than allocating sensors independently to each target, the system enters conflict resolution mode to minimize the maximum team cost across all competing targets.

Let $\mathcal{T} = \{k_1 k_2 \dots k_n\}$ denote the targets requiring handovers, and $S_{\text{available}} = \{i_1 i_2 \dots i_m\}$ denote available sensors (excluding active trackers). For each assignment $A: \mathcal{T} \rightarrow S_{\text{available}} \times S_{\text{available}}$ mapping targets to non-overlapping pairs of sensors, we determine:

$$\text{cost}(A) = \max_{k \in \mathcal{T}} \sum_{i \in A(k)} \text{bid}_i(k) \quad (16)$$

where $\text{bid}_i(k)$ is Sensor i 's bid for Target k 's handover mission and $A(k)$ denotes the two-sensor team assigned to Target k . The optimal assignment $A^* = \arg \min_A \text{cost}(A)$ minimizes the worst-case team cost, ensuring no target is disproportionately disadvantaged in resource allocation.

Assignment determination enumerates valid non-overlapping sensor team combinations. For two-target conflicts, for example, all possible pairs ($\{s_1 s_2\} \{s_3 s_4\}$) where $s_i \in S_{\text{available}}$ and teams share no common sensors are evaluated to find the min-max solution. To reduce computational complexity from $O(|S_{\text{available}}|^4)$ exhaustive search, our strategy speeds-up the search by considering only the top- k bidders per target, reducing complexity to $O(k^4)$

where $k \ll |S_{\text{available}}|$. This approximation maintains near-optimal solutions while enabling real-time conflict resolution for $|\mathcal{T}| = 2$ simultaneous calls. When no conflicts exist, sensors are assigned independently based on lowest individual bids without minimax optimization.

G. Information Sharing During Tracking

While handover calls are broadcast globally, target state information is shared locally between sensors within communication range r_c . Each tracking sensor periodically broadcasts its EKF state estimate:

$$\hat{\mathbf{x}}_k = [\hat{x}_k \hat{y}_k \hat{v}_{kx} \hat{v}_{ky}]^T \quad (17)$$

to neighbors. Receiving sensors compute a confidence-weighted combination of estimates from all trackers within range:

$$\hat{\mathbf{x}}_{\text{shared}} = \frac{\sum_{j \in \mathcal{N}_i} w_j \hat{\mathbf{x}}_j}{\sum_{j \in \mathcal{N}_i} w_j} \quad w_j = \frac{e^{-0.1 \Delta t_j}}{1 + 0.1 d_{ij}} \quad (18)$$

where $\mathcal{N}_i$ denotes sensors within communication range of Sensor i , Δt_j is the time since Sensor j 's last detection, and d_{ij} is the inter-sensor distance. This fusion provides confidence-degraded estimates as information ages or distance increases, enabling auxiliary sensors to use shared target information for guidance without requiring direct detection.

Each repositioning sensor determines a confidence metric $\alpha_i \in [0, 1]$ for shared information based on the aggregate weight:

$$\alpha_i = \frac{\sum_{j \in \mathcal{N}_i} w_j}{\max(\sum_{j \in \mathcal{N}_i} w_j)} \quad (19)$$

This confidence metric is used in the bidding mechanism (Section IV.E, penalty component 4) to penalize assignments when available information is stale or spatially distant.

H. Adaptive Guidance for Sensor Repositioning

Upon winning the handover bid, idle sensors transition to *REPOSITIONING* state to reach the predicted handover point $\mathbf{p}_{\text{handover}}$ calculated in Section IV.D (addressing Sub-problem C.2). The sensor employs a distance-dependent adaptive guidance strategy.

(1) Phase 1: Adaptive Guidance ($> \delta_{\text{switch}}$ units)

When far from the handover point (distance $> \delta_{\text{switch}}$ from $\mathbf{p}_{\text{handover}}$), sensors utilize shared target information (Section IV.G) from neighboring trackers within communication range, computing confidence $\alpha \in [0, 1]$ based on information age and source proximity. Three scenarios arise:

- *Low Confidence* ($\alpha < \alpha_{\text{threshold}}$): The sensor aborts repositioning and transitions to *RETURNING_HOME*, as unreliable target information renders the predicted handover point invalid.
- *Target Not Yet Passed*: When geometric analysis indicates the target has not moved beyond the predicted

handover point ($\alpha \geq \alpha_{\text{threshold}}$), the sensor navigates directly toward $\mathbf{p}_{\text{handover}}$ using:

$$\mathbf{v}_{\text{command}} = v_s \frac{\mathbf{p}_{\text{handover}} - \mathbf{s}_i}{\|\mathbf{p}_{\text{handover}} - \mathbf{s}_i\|} \quad (20)$$

- *Target Passed Handover Point:* When geometric analysis indicates the target has moved beyond the predicted handover point and $\alpha \geq \alpha_{\text{threshold}}$, the sensor employs a navigational strategy to adapt its trajectory toward the target's current position. Herein, proportional navigation guidance (PNG) is employed. PNG is a guidance law that commands the pursuer to rotate its velocity vector at a rate proportional to the line-of-sight (LOS) rotation rate [55].

The PNG acceleration command is:

$$a_{\text{PNG}} = N' \dot{\lambda} V_c, \quad (21)$$

where a_{PNG} is the commanded lateral acceleration (perpendicular to the sensor's velocity vector), N' is the navigation constant, $\dot{\lambda}$ is the LOS rate, and V_c is the closing velocity.

PNG ensures a collision course by nullifying the LOS rate and minimizing time-to-handover. The navigation constant $N' = 3$ provides near-optimal interception trajectories for constant-velocity targets [57].

(2) Phase 2: Final Approach ($\leq \delta_{\text{switch}}$ units)

When within δ_{switch} units of $\mathbf{p}_{\text{handover}}$ and the target has not yet been acquired, sensors revert to direct navigation regardless of shared information (same guidance as (20)).

Upon reaching $\mathbf{p}_{\text{handover}}$ (arrival), if target acquisition is successful, they transition to *TRACKING* or *RETURNING_HOME*, if acquisition fails.

I. Dual-Tracker Configuration and Immediate Handover

Herein, it is proposed that the Tracking Coordinator maintains up to two active trackers per target for robustness against individual sensor failure or detection loss. When a new sensor detects a target already being tracked, the Tracking Coordinator evaluates whether to accept the new tracker based on current tracker count.

If the target already has two active trackers, the system identifies the tracker that is farthest from its home position:

$$i_{\text{release}} = \arg \max_{i \in \mathcal{T}_k} \|\mathbf{s}_i - \mathbf{h}_i\| \quad (22)$$

where $\mathcal{T}_k$ denotes the set of sensors currently tracking target k . The tracker with maximum home displacement i_{release} is released and transitions to *RETURNING_HOME*, allowing the new detecting sensor to assume tracking responsibility (immediate handover). This policy balances tracking redundancy with network topology preservation.

Upon assignment, each new tracker initializes its EKF state estimate for the target using the current target position measurement, with initial state $\hat{\mathbf{x}}_k = [\mathbf{z}_{ik} \mathbf{0}]^T$ (zero initial

velocity) and large initial covariance to reflect velocity uncertainty.

When a target exits the monitored region (travels beyond all sensor ranges), assigned trackers transition to *RETURNING_HOME* and remove the target from their tracking list.

J. EKF-Based Iso-Probability Search Contour Generation

Upon confirmed target loss, the last tracking sensor generates search regions based on its final EKF state estimate. These regions are represented as iso-probability curves of equal probability density that define areas where the target is equally likely to be found. Traditional iso-probability curve methods compute full probability density distributions over discretized grids requiring centralized offline computation [4], [56], [57]. In contrast, the proposed lightweight approach leverages the existing EKF covariance matrix directly to generate these contours without additional computational overhead.

The position covariance submatrix $\mathbf{P}_{\text{pos}} = \mathbf{P}_k(1:21:2)$ defines a 2D Gaussian distribution over target location. Eigenvalue decomposition $\mathbf{P}_{\text{pos}} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$ yields principal axes directions (columns of $\mathbf{V}$) and uncertainty magnitudes (diagonal of $\mathbf{\Lambda} = \text{diag}(\lambda_1 \lambda_2)$). For confidence level α , the corresponding chi-square value $\chi_\alpha^2(2)$ scales the uncertainty ellipse. The contour parameterization is:

$$\mathbf{c}(\theta) = [\hat{x}_k \hat{y}_k] + \sqrt{\chi_\alpha^2(2)} \mathbf{V} [\sqrt{\lambda_1} \cos \theta \sqrt{\lambda_2} \sin \theta]^T \quad (23)$$

for $\theta \in [0, 2\pi]$. For a 2D Gaussian distribution, n-sigma ellipses correspond to chi-squared values with two degrees of freedom, yielding confidence levels of 39.35% for 1σ , 86.47% for 2σ , and 98.89% for 3σ , Fig. 7.

These nested elliptical contours represent increasingly likely target locations and provide spatial search regions for subsequent search operations, though full search algorithm implementation is reserved for future work.

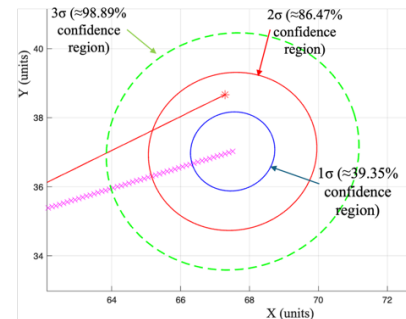


Fig. 7 EKF-based probabilistic search contours generated after target loss.

V. SIMULATED EXPERIMENTS

A. Simulation Setup

Numerous simulations were conducted to evaluate the proposed framework's performance in multi-target tracking scenarios with varying target velocities and simultaneous handover demands. Two representative scenarios, included herein, demonstrate key system capabilities: (1) conflict resolution when multiple targets simultaneously require auxiliary sensors, and (2) scalable coordination among three targets with progressive velocities and staggered entry times. Performance is evaluated through tracking coverage duration, sensor engagement rates, handover success rates, and network topology preservation.

The DES simulation assumes availability of long-range communication infrastructure enabling global message broadcasting for handover coordination. This architectural choice ensures all sensors can participate in bidding regardless of network position, maximizing auxiliary sensor selection quality.

The WSN employs $N = 25$ sensors in a 5×5 hexagonal grid with inter-node spacing $d = 12$ units and detection radius $r_d = 15$ units. Sensor velocity is $v_s = 0.75$ units/time unit, and communication range $r_c = 18$ units enables each sensor to coordinate with first and second-tier neighbors. The hexagonal geometry provides optimal coverage while maintaining structured communication patterns for distributed coordination.

EKF parameters include process noise covariance $\mathbf{Q} = \text{diag}(0.01, 0.01)$ and measurement noise covariance $\mathbf{R} = \text{diag}(0.1, 0.1)$, adopted from [58]. Velocity convergence is detected when variance components satisfy $\mathbf{P}_k(33) < \sigma_{v_{\text{thresh}}}^2 = 0.006$, and $\mathbf{P}_k(44) < \sigma_{v_{\text{thresh}}}^2 = 0.006$, typically achieved within 1 time unit of continuous tracking. This threshold was determined through sensitivity analysis, representing the point where velocity estimates achieve sufficient accuracy for reliable loss prediction.

Bidding weights are $w_n = 0.4$, $w_s = 0.3$, $w_t = 0.2$, $w_u = 0.1$ unless otherwise stated. These weights prioritize network topology preservation ($w_n = 0.4$) to ensure long-term operational sustainability, followed by spatial efficiency ($w_s = 0.3$) to minimize travel distance, temporal feasibility ($w_t = 0.2$) to ensure timely arrival, and information quality ($w_u = 0.1$) to account for tracking uncertainty. This allocation reflects the dual objectives stated in Section III.B: maximizing tracking coverage while minimizing displacement from home positions.

B. Example 1: Conflict Resolution Under Simultaneous Handover Calls

Fig. 8 illustrates the system's conflict resolution mechanism when multiple sensors require handover resources. At $t = 571 - 572$ time units, two tracking

sensors detect imminent target loss through EKF-based prediction and issue handover calls within the conflict detection window. This triggers the DES conflict resolution mode, which processes the competing requests to allocate non-overlapping sensor teams.

The min-max algorithm processes the competing requests at $t = 573 - 576$ time unit. Rather than evaluating all possible sensor-target assignments, the algorithm limits the search space to the top- k bidders per target (typically $k = 8$) based on initial bid calculations. This constraint reduces computational complexity from $O(|S_{\text{available}}|^4)$ exhaustive search to $O(k^4)$ where $k \ll |S_{\text{available}}|$, enabling real-time conflict resolution while maintaining near-optimal assignment quality.

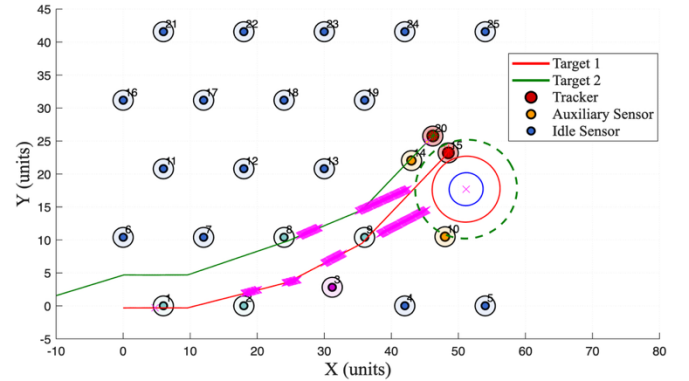


Fig. 8 Conflict resolution demonstration.

Fig. 9 shows the post-resolution assignments at $t = 576$ time unit. The algorithm successfully allocated non-overlapping handover sensor pairs: Sensors 10 and 14 are assigned as auxiliary sensors for Target 1, while Sensors 19 and 25 are assigned to Target 2. The min-max objective ensures balanced resource allocation by minimizing the maximum team cost across both targets, preventing one target from receiving disproportionately suboptimal auxiliary sensors. Both auxiliary sensor pairs successfully complete their missions, with handovers occurring at $t = 634$ time unit (Target 1) and $t = 641$ time unit (Target 2), maintaining continuous tracking throughout the conflict resolution process.

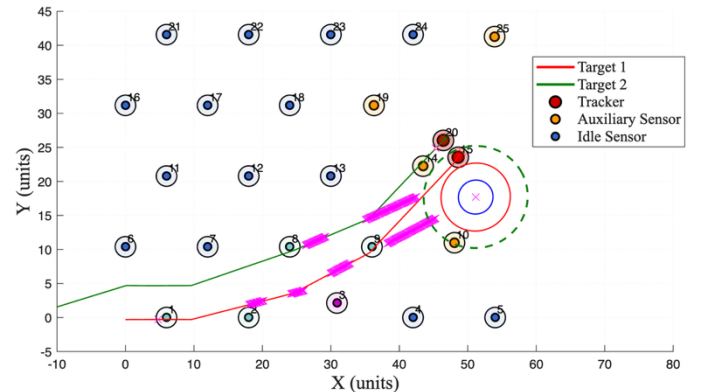


Fig. 9 Post-resolution assignments at $t=57.6$ time units.

C. Example 2: Three-Target Coordination with Progressive Velocities

Fig. 10 shows system scalability through coordination of three targets with progressively increasing velocities and staggered entry times. Target 1 (red trajectory) enters at $t = 0$ time unit with velocity 10 unit/time unit, Target 2 (green trajectory) enters at $t = 40$ time unit with velocity 11 unit/time unit, and Target 3 (blue trajectory) enters at $t = 65$ time unit with velocity 12 unit/time unit. This configuration tests the framework's ability to maintain tracking performance under increasing speed demands and temporal complexity arising from overlapping target trajectories and concurrent tracking operations.

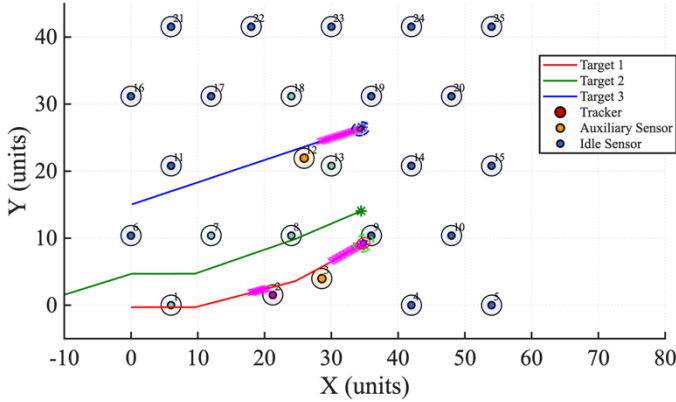


Fig. 10 Three-target tracking with progressive velocity increases.

The system successfully maintained dual-tracker configurations for all three targets throughout the simulation despite overlapping sensor coverage areas. Distributed EKF estimation converged for all targets within one time unit of initial detection, with final velocity variance components satisfying $\mathbf{P}_k(33) < 0006$ and $\mathbf{P}_k(44) < 0006$. The converged estimates enabled stable loss predictions, triggering handover calls with sufficient lead time for successful handovers even at the highest target velocity of 12 unit/time unit.

Fig. 11 demonstrates the PNG behavior for a single repositioning mission extracted from the three-target scenario. Sensor 25, initially at its home position, receives a handover call for Target 3 (velocity 12 unit/time unit) and wins the bidding process. The trajectory visualization shows three distinct phases:

- *Direct Pursuit Phase:* Upon transitioning to REPOSITIONING state, the sensor initially moves directly toward the safe handover point using the guidance law from Section IV.H, Phase 1.
- *PNG Engagement:* As shared information indicates the target has passed the predicted handover point, the sensor activates PNG guidance (Section IV.H). The trajectory curves show the characteristic smooth pursuit path as PNG nullifies the line-of-sight rate, adapting to the target's actual motion rather than

rigidly following the initial prediction.

- *Final Approach and Return:* The sensor reaches the vicinity of the handover point but fails to reacquire the target, due to insufficient information confidence (shared target state became stale or unreliable). Following the protocol in Section IV.H, the sensor transitions to *RETURNING_HOME* state (blue "Return" trajectory), abandoning the handover mission and returning to its home position to preserve network topology.

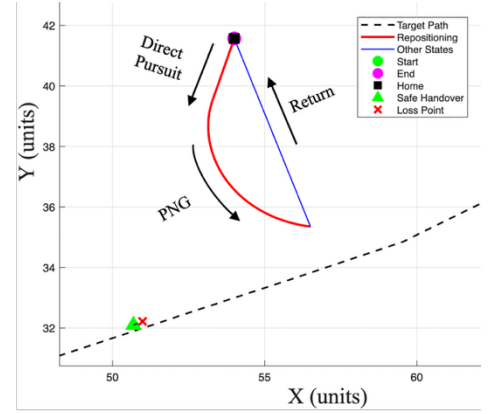


Fig. 11 Sensor 25 trajectory during handover mission for Target 3.

The curved pursuit trajectory shows PNG's ability to compensate for prediction errors and target motion deviations, providing graceful degradation when initial handover point predictions prove inaccurate. This adaptive guidance capability contributed to the 765% reacquisition success rate among mobilized sensors in the three-target scenario.

Eight successful handovers were executed across the three-target scenario. Of the 25 sensors, 17 (68%) were mobilized for repositioning missions: 13 sensors successfully reacquired targets at predicted handover points, while 4 sensors reached handover points but failed target reacquisition due to target trajectory deviations. This 76.5% success rate among mobilized sensors shows effective predictive accuracy while highlighting the inherent uncertainty in forecasting fast-moving target positions. The four unsuccessful repositioning attempts transitioned to *RETURNING_HOME* state, preserving network topology without compromising ongoing tracking operations.

This high utilization rate demonstrates effective distributed coordination, with sensors dynamically transitioning between *IDLE*, *TRACKING*, and *REPOSITIONING* states based on bidding outcomes and tracker assignments. No sensor exceeded the displacement constraint $d_{max} = d = 12$ units from its home position, confirming that the network preservation penalty ($w_n = 04$) successfully balanced mission effectiveness with topology maintenance.

D. Performance Summary

The two scenarios collectively demonstrated four key system capabilities. First, the conflict resolution mechanism successfully manages simultaneous multi-target demands through min-max optimization. Second, the dual-tracker configuration provides robustness against individual sensor failures or transient detection losses, as evidenced by continuous tracking despite overlapping coverage areas and high sensor utilization. Third, the predictive handover framework scales upward effectively while maintaining tracking continuity under progressive velocity increases. Fourth, the bidding mechanism's network preservation component ensures sensors remain within displacement constraints, preserving network topology for subsequent tracking missions.

VI. CONCLUSIONS

This paper presents a distributed coordination framework for multi-target tracking in wireless sensor networks based on DES principles and predictive guidance theory. The proposed approach transforms passive tracking, where sensors react to target loss events, into active swarm coordination where sensors proactively position themselves at predicted handover points before current trackers lose detection. Results from simulated experiments validate the effectiveness of predictive handover, demonstrating 122% improvement in tracking coverage (80.4 vs 36.2 time units) and 117% increase in sensor engagement (52% vs 24% network utilization) compared to reactive tracking baselines. The system achieved 10 successful predictive handovers from 13 handover calls (76.9% success rate), maintaining continuous multi-target tracking through proactive sensor repositioning. Active coordination engaged 28 percentage points more of the sensor network and doubled detection events, enabling persistent tracking across trajectory discontinuities that caused complete tracking loss in reactive approaches.

The integration of predictive handover, competitive bidding, and conflict resolution enables proactive sensor coordination while maintaining network topology. Simulation results demonstrate substantial improvements in tracking coverage and sensor engagement compared to reactive approaches, validating the framework's effectiveness across diverse target scenarios.

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